

Absurdly Accurate Clock

Pico 2 / RP2350 pin-assignment reference

REV 2 | REFRESHED 26 SEP 2026

US LETTER - LANDSCAPE | 1 / 1

Assembly: Populate both 20-pin header rows. Top/component view, USB at top: pin 1 upper left; pin 20 lower left; pin 21 lower right; pin 40 upper right. Physical numbers below are board header pins, not chip pins.

Physical	GPIO	Direction*	Peripheral / function	Target / signal	Wiring / implementation notes
1	GP0	OUT	UART0 TX (F2)	GPS RX	Pico TX to GPS RX; future configuration connection. Current firmware sends no receiver commands.
2	GP1	IN	UART0 RX (F2)	GPS TX	NMEA / UBX serial data from GPS.
4	GP2	IN	GPIO / SIO (F5)	GPS PPS	Rising-edge GPS PPS input.
6	GP4	OUT	UART1 TX (F2)	MAX3232 T1IN	T1OUT (RS-232) to Posiflex PD-2200 RX.
7	GP5	IN / reserve	UART1 RX (F2)	MAX3232 R1OUT	Reserved; UART RX disabled. Optional future return path.
9	GP6	IN	GPIO / SIO (F5)	UI button	Active-low to GND; internal pull-up; 30 ms debounce. Release before rearm; no held-button repeat.
11	GP8	IN	SPI1 RX (F1)	ESP32 MOSI	SPI data from ESP32 to Pico.
12	GP9	IN	SPI1 CSn (F1)	ESP32 CS	Active-low chip select from ESP32.
14	GP10	IN	SPI1 SCK (F1)	ESP32 SCLK	SPI clock from ESP32.
15	GP11	OUT	SPI1 TX (F1)	ESP32 MISO	SPI data from Pico to ESP32.
16	GP12	OUT	GPIO / SIO (F5)	ESP32 TIME_SYNC	Qualified, software-delayed sync for an accepted authoritative UTC/PPS boundary; Pico to ESP32. See protocol v1 below.
17	GP13	TBD / reserve	GPIO / SIO (F5)	ESP32 control / IRQ	Reserved; direction and polarity not yet assigned.

*Directions are relative to the Pico. ESP32 is the SPI controller; Pico/RP2350 is the native SPI1 peripheral. ESP32 GPIO numbers remain TBD; NTP service is future ESP32 work.

Protocol v1 / TIME_SYNC: docs/clock-network-protocol.md defines qualification and delay reporting. TIME_SYNC is not raw PPS, a zero-delay signal, or a calibrated precision output.

Keep free for expansion: GP14 / pin 19; GP15 / 20; GP16 / 21; GP17 / 22; GP18 / 24; GP19 / 25; GP20 / 26; GP21 / 27; GP22 / 29. Also unassigned: GP3 / 5, GP7 / 10, GP26 / 31, GP27 / 32, GP28 / 34.

Ground and interface wiring: Nearby GND pins: 3, 8, 13, 18; other digital GND pins: 23, 28, 38 (pin 33 is AGND). Shared ground is required; use 3.3 V logic at the Pico. Keep RS-232 on the display side of the MAX3232; verify module logic levels and PD-2200 connector wiring before assembly. Power distribution is not specified by this sheet.

Pinmux verified: UART0, UART1 and SPI1 assignments coexist. GP12 / GP13 stay GPIO, despite their other alternate functions. In C++ / PlatformIO, assign GPIO numbers explicitly; do not rely on framework defaults. F1 = SPI, F2 = UART, F5 = SIO. Leave unused reserved signals undriven until their interface is defined.

Authoritative references checked 22 Sep 2026: Raspberry Pi Pico 2 pinout (board numbering) and RP2350 Datasheet, section 1.2.3, Table 3 (GPIO mux). Documentation refreshed against current source and protocol on 26 Sep 2026; electrical Rev 2 unchanged. No new bench qualification performed.

Pico 2 pinout: <https://datasheets.raspberrypi.com/pico/Pico-2-Pinout.pdf>

RP2350 datasheet: <https://datasheets.raspberrypi.com/rp2350/rp2350-datasheet.pdf>